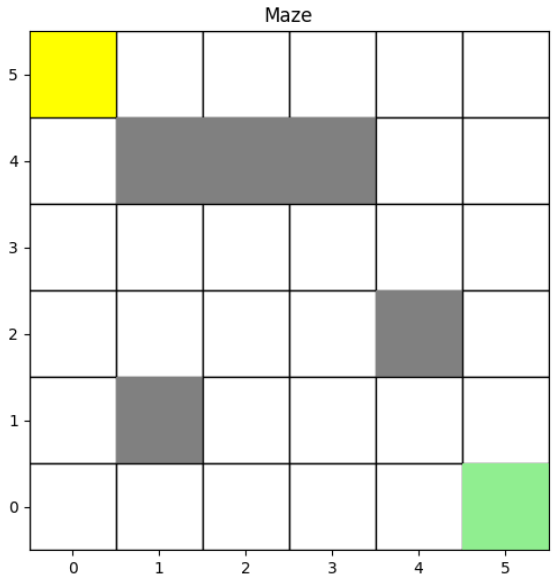
Imtiyaz Mohammed **Reinforcement Learning**

Z23749265 **Project - 01**

Q-LEARNING ALGORITHM

Co-lab Link:
https://colab.research.google.com/drive/1k6uXuxfFGVweHlTdif4baQgD6yRbzFsA#scrollTo=TLmxuwWaTq3v

Results:



Maze

1. **Maze Description:**
   - A 6x6 grid world was created, with several obstacles (grey cells) and a goal (light green cell) positioned at (5, 5). The start position (yellow cell) was set at (0, 0).
   - Walls were strategically placed to challenge the agent's pathfinding, forcing it to explore and avoid these obstacles.
   - Maze has 36 states and four actions (Up, Down, Left Right).
   - Rewards is 10 to reach the goal, -10 for hitting the wall and -1 for step penalty
2. **Q-learning:**
   - The Q-learning algorithm was run for 500 episodes, with a learning rate of 0.1, a discount factor of 0.9, and an epsilon-greedy policy for exploration ($\varepsilon = 0.1$).
   - As expected, initially, the agent explored the environment and randomly encountered the goal after many steps. Over time, the algorithm learned more efficient paths.

$$Q(s, a) \leftarrow Q(s, a) + \alpha (r + \gamma * \text{Max } Q(s', a') - Q(s, a))$$

s= current state,

a= current action,

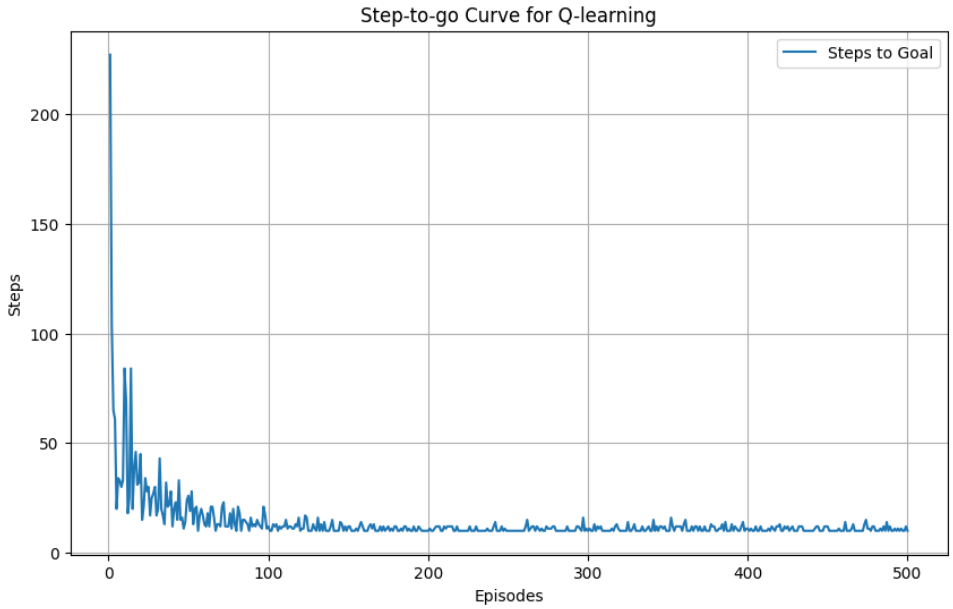
s'= next state,

a'= next action,

r= reward obtained after acting **a** in state **s,**

α= learning rate,

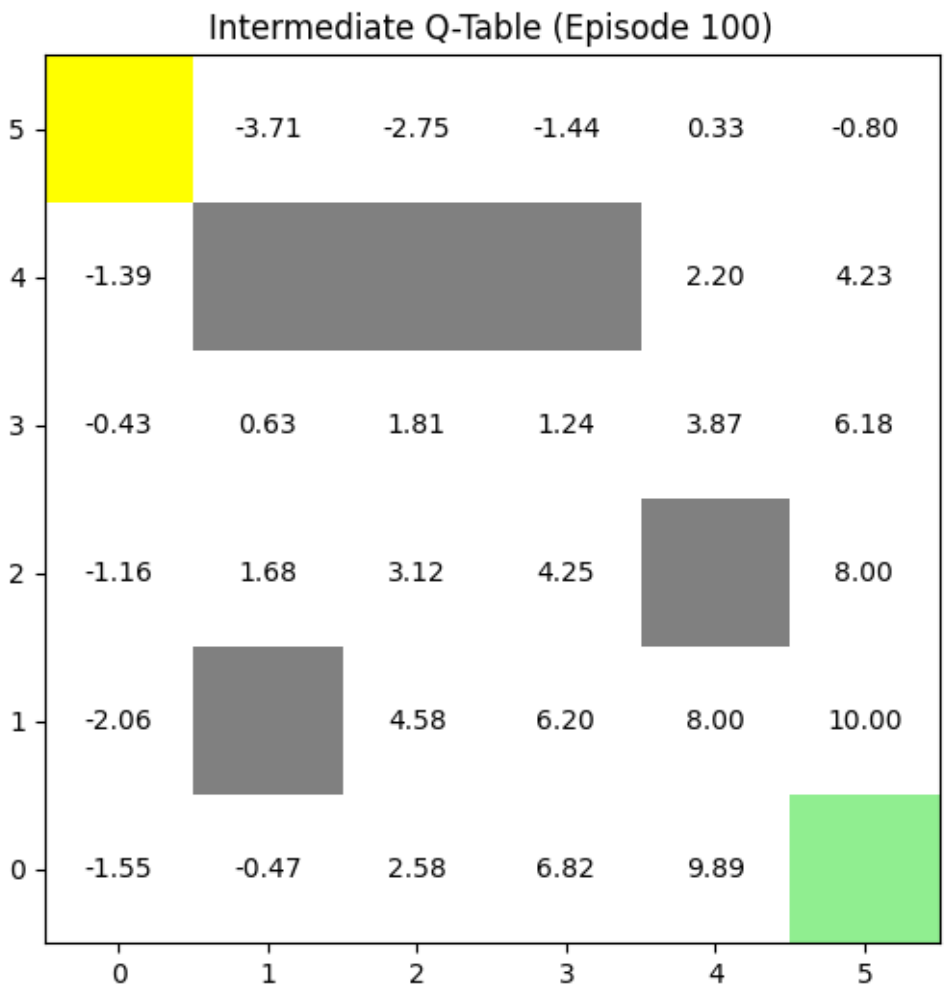
Γ= discount factor.



Step-to-Go Curve

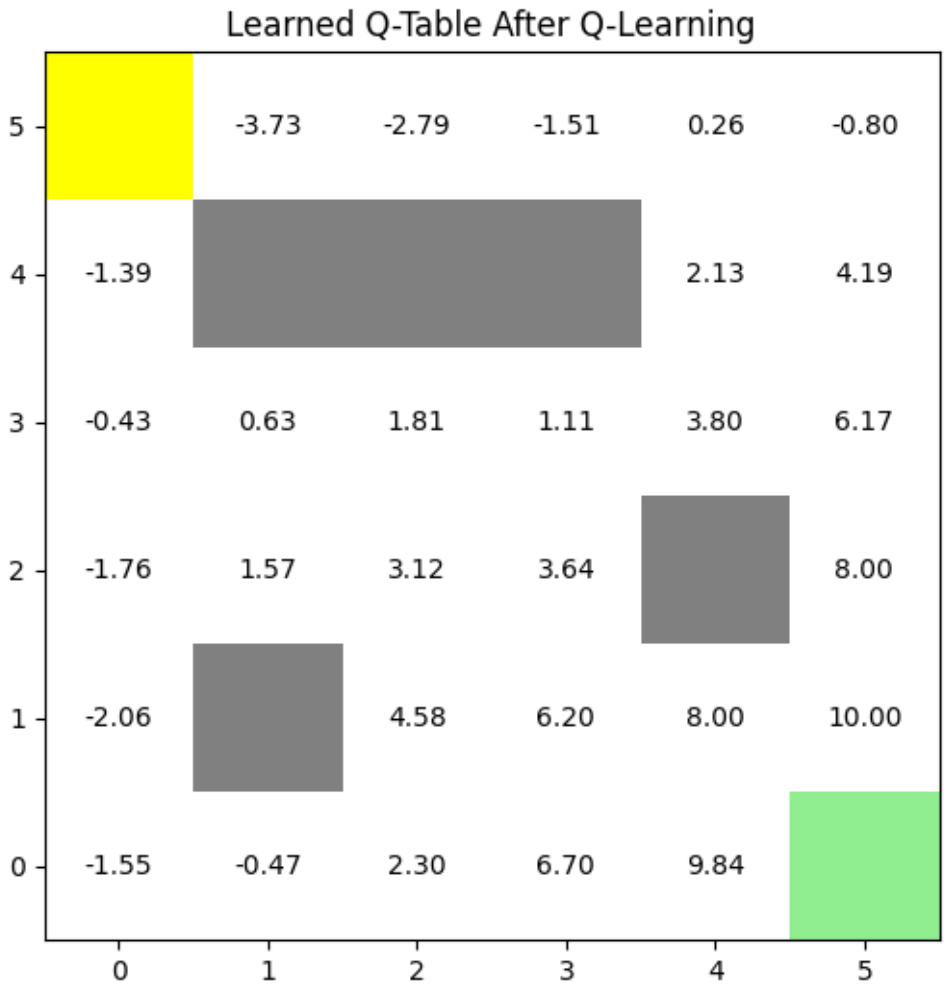
3. Step-to-Go Curve:

- The **step-to-go curve** demonstrates a clear improvement in the agent's ability to find the goal over time.
- In the early episodes, the agent takes many steps to reach the goal, primarily because it explores the grid without knowledge of the best path. However, as episodes progress, the number of steps required to reach the goal decreases significantly. The agent learns to find shorter, more optimal paths.
- The curve levels off towards the end, indicating that the agent has learned the optimal policy and consistently follows the shortest path to the goal.



4. **Intermediate Q-Table (Episode 100):**

- The **intermediate Q-table** at episode 100 shows partially learned values. At this point, the agent has discovered some efficient paths but still exhibits uncertainty in several areas, especially when near the walls or obstacles.
- Q-values for cells near walls were generally low, indicating that the agent has learned to avoid these states. However, the Q-values for cells closer to the goal are higher, reflecting the agent's growing understanding that these states lead to successful outcomes.



5. **Learned Q-Table (After 500 Episodes):**

- The **final Q-table** demonstrates a fully learned policy. The Q-values near the goal state are maximized, showing that the agent consistently chooses actions that bring it closer to the goal.
- Q-values in states farther from the goal and near walls have very low values, meaning the agent has learned to avoid them to minimize penalties.
- The agent follows the optimal policy by consistently selecting actions that maximize its cumulative reward, thus reaching the goal in the fewest possible steps.

6. **Analysis:**

- In the beginning, the agent explores the grid heavily, but as it gains more knowledge through experience, it exploits its learned values to find the best paths efficiently.
- The combination of exploration with the epsilon-greedy policy and a discount factor that prioritizes future rewards allowed the agent to learn an optimal policy effectively.
- The placement of walls creates a challenge for the agent. The algorithm successfully learns to navigate around these obstacles, avoiding states that lead to negative rewards.
- More sophisticated reinforcement learning algorithms (such as Deep Q-Networks or Policy Gradient Methods) could be applied to handle larger, continuous, or stochastic environments.

Summary:

The agent successfully learned an optimal path in the 6x6 grid world using Q-learning, overcoming obstacles, and reducing the number of steps to reach the goal as training progressed. The step-to-go curve and Q-table analysis provide clear evidence of the agent's improvement over time.